Probability Review 2

Continuous Random Variables

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Roadmap

1. From Counts to Measurements
2. Continuous Random Variables
3. The Probability Density Function
4. The Cumulative Distribution Function
5. Expectation and Variance for Continuous RVs
6. The Uniform Distribution
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1. From Counts to Measurements

1.1 From Counts to Measurements

Lecture 1 handled discrete random variables — outcomes you can *count*: coin tosses, dice rolls, number of heads, number of customers.

But many real economic quantities are measured on a continuum:

- Height (1.732 m, 1.7321 m, anything in between . . .).
- Time (delivery arrives at 2:17:33.482 pm).
- Price, stock return, GPA, wait time, weight.

These variables don't have a finite or countable list of outcomes — between any two values there are infinitely many more. The PMF $p_X(x) = \mathbb{P}(X = x)$ from Lecture 1 no longer works, and we need a new tool.

The new tool is a density curve, and probabilities will be areas under the curve. The rest of this lecture makes that precise — but you should hold that picture in mind from the start.

2. Continuous Random Variables

2.1 Motivating Example: A Delivery Window

Setup. A delivery company tells you your new refrigerator will arrive sometime between 2:00 pm and 3:00 pm. Let X be the exact arrival time, measured in minutes after 2:00 pm. So $X \in [0, 60]$.

Two questions:

- What is $\mathbb{P}(X = 17.33482\ldots)$? Vanishingly small — one specific instant out of infinitely many.
- What is $\mathbb{P}(X \leq 20)$? Meaningful — a one-third chunk of the hour, so $1/3$.

Takeaway. For a continuous variable, we will talk about probabilities of *ranges* ($X \in [a, b]$), not probabilities of *individual values*.

2.2 Continuous Random Variable

Definition
A random variable is a function

$$X : S \longrightarrow \mathbb{R}$$

that assigns a real number to every outcome of the random experiment (recall Lecture 1.6.1). We call X continuous if its range is an *interval* of $\mathbb{R}$, so the values X takes form a continuum rather than a finite or countable list.

Notation reminder. A function $f : A \to B$ is a rule that takes each input from A and returns exactly one output in B. "$X : S \to \mathbb{R}$" says: feed me an outcome s from the sample space, and I give you back a real number $X(s)$. Many objects in this course are functions — densities, CDFs, the

2.2 Continuous Random Variable 2

regression function $f(x)$ in §10, loss functions later — so keep the idea in mind.

Contrast with the discrete case from Lecture 1.6:

- **Discrete** X: range is a finite/countable subset of $\mathbb{R}$ (e.g. $\{0, 1, 2\}$). Described by a PMF.
- **Continuous** X: range is an interval of $\mathbb{R}$ (e.g. $[0, 60]$). Described by a density (next section).

2.3 The Measure-Zero Puzzle

Key fact
For a continuous random variable X,

$$\mathbb{P}(X = x) = 0 \quad \text{for every fixed value } x.$$

Why? There are infinitely many possible values in any interval, but probabilities must total 1. If any single point had positive probability, summing across infinitely many such points would exceed 1. So each individual point must get probability zero.

Consequence. For continuous X, the endpoints of an interval don't matter:

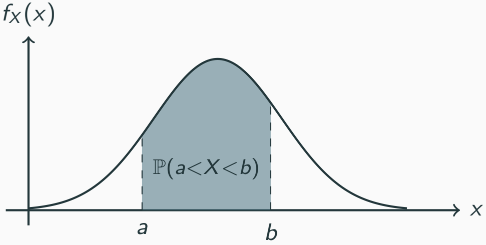
$$\mathbb{P}(a < X < b) = \mathbb{P}(a \leq X \leq b).$$

Including or excluding the boundary contributes zero.

3. The Probability Density Function

3.1 Probability as Area Under a Curve

For a continuous random variable, probabilities are described by a smooth curve $f_X(x)$ called the probability density function (PDF). The probability that X falls in any range $[a, b]$ is the *area under that curve* between a and b.



Key insight. The *height* $f_X(x)$ is not a probability; the shaded *area* is.

3.2 Formal Definition of the PDF

Definition
The probability density function of a continuous random variable X is a function $f_X(x)$ satisfying:

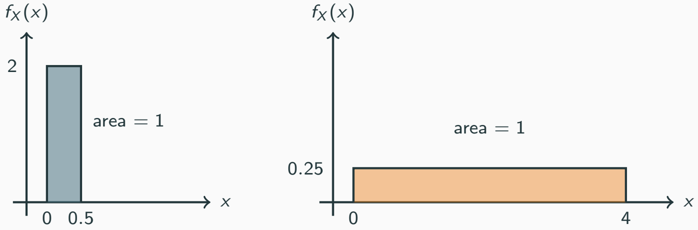
- $f_X(x) \geq 0$ for all x,
- the total area under the curve equals 1: $\displaystyle\int_{-\infty}^{\infty} f_X(x)\,dx = 1$,
- for any $a \leq b$, $\mathbb{P}(a \leq X \leq b) = \displaystyle\int_a^b f_X(x)\,dx$ (the area under f_X between a and b).

Notation reminder. The symbol $\int_a^b f_X(x)\,dx$ is shorthand for the area under the curve f_X between a and b.

In this course we will never compute an integral. We will read areas off a picture, or we will let software compute them for us. The integral notation is here so the formulas look right — not because you must evaluate them.

3.3 Density Is Not Probability

A common confusion: the height $f_X(x)$ of the density curve *can exceed* 1. Density is not probability — only area is.



Both rectangles have total area 1 — they are valid densities. The left one has *height* 2 (greater than 1!) over a narrow base; the right one has height 0.25 over a wide base. What matters is area, not height.

3.4 Worked Example: Reading a Uniform Density

Setup. Let X have density $f_X(x) = 1/10$ on $[2, 12]$ (and 0 elsewhere). We'll meet this distribution properly in §6.

Step 1: draw it. A rectangle from $x = 2$ to $x = 12$ with height $1/10$. Total area $= 10 \times \frac{1}{10} = 1$. ✓

Step 2: read off probabilities by area.

$$\mathbb{P}(2 \le X \le 10) = \underbrace{8}_{\text{base}} \times \underbrace{\tfrac{1}{10}}_{\text{height}} = 0.8.$$

No integration needed — it's the area of a rectangle.

Step 3: single points contribute zero. $\mathbb{P}(X = 3) = 0$, $\mathbb{P}(X \le 2) = 0$.

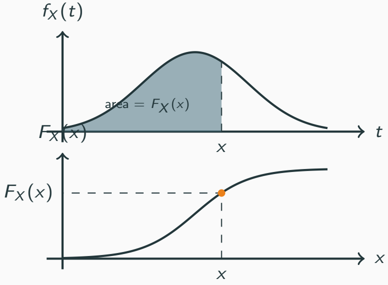
Interpretation. Probabilities are areas you can see.

4. The Cumulative Distribution Function

Definition
The cumulative distribution function (CDF) of X is

$$F_X(x) = \mathbb{P}(X \le x).$$

Geometrically: the area under f_X from $-\infty$ up to x — the "running total" of probability.



4.2 PDF $\leftrightarrow$ CDF Relationship

Key result
For any $a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a).$$

Why? The area between a and b equals the area up to b minus the area up to a: take the whole shaded region under f_X from $-\infty$ to b and subtract the part from $-\infty$ to a. The strip between a and b is what's left.

Two basic properties:

- $F_X(x)$ is non-decreasing: more area can only get bigger.
- $F_X(-\infty) = 0$ and $F_X(\infty) = 1$.

Intuition. The PDF is the "rate of climb" of the CDF (calculus aside: $f_X = F_X'$). You won't compute this; the picture is the point.

4.3 Worked Example: Using a CDF

Setup. Let $X \sim U[0, 10]$ with CDF $F_X(x) = x/10$ on $[0, 10]$.

Step 1: use the formula.

$$\mathbb{P}(3 \leq X \leq 7) = F_X(7) - F_X(3) = \tfrac{7}{10} - \tfrac{3}{10} = 0.4.$$

Step 2: sanity check via area. The PDF is a rectangle of height $1/10$ on $[0, 10]$. Between $x = 3$ and $x = 7$ the strip has width 4 and height $1/10$, so area $= 4 \times \frac{1}{10} = 0.4$. ✓

Interpretation. The CDF gives running totals; subtract to get interval probabilities. This is exactly the move we'll use to read Normal probabilities in §8.

5. Expectation and Variance for Continuous RVs

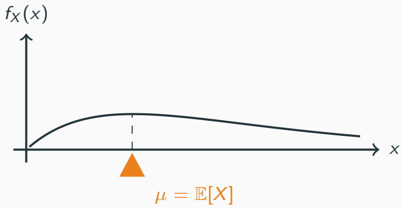
5.1 Expectation: Same Idea, New Object

Definition
The expected value of a continuous random variable X is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\, f_X(x)\, dx.$$

Notation reminder. This integral is the continuous analogue of the sum $\sum_x x\, p_X(x)$ from Lecture 1.7.1. The summation has become an "infinite weighted sum" over all x, weighted by the density.

Geometric intuition. $\mathbb{E}[X]$ is the balance point of the density: the point where the curve would balance on a fulcrum. For a symmetric density, that's the center. For a right-skewed density, the balance point is pulled to the right.



5.2 Variance: Same Definitions Carry Over

Definition

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \qquad \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

All the rules from Lecture 1 transfer unchanged:

- Linearity of expectation: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ (Lecture 1.7.2).
- Computational variance formula: $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ (Lecture 1.7.4).
- Affine scaling: $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$ (Lecture 1.7.5).
- Independent sum: if $X \perp\!\!\!\perp Y$ then $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Where this shows up later. These properties drive the standard-error formulas for OLS coefficient estimates (Lecture 7).

5.3 Independence of Random Variables

In Lecture 1.4.5 we defined what it means for two *events* to be independent. We now extend the idea to two *random variables* — since the variance rules in the next two frames depend on it.

Definition (intuitive)
Two random variables X and Y are independent, written $X \perp\!\!\!\perp Y$, if knowing the value of one tells you *nothing* about the distribution of the other.

Notation reminder. $X \perp\!\!\!\perp Y$ reads "X and Y are independent."

Formally, for every pair of values x, y:

$$\mathbb{P}(X \leq x,\ Y \leq y) \ = \ \mathbb{P}(X \leq x)\,\mathbb{P}(Y \leq y).$$

The joint probability factors into a product of the individual probabilities. (Equivalently: the joint PMF/PDF factorizes, $p_{X,Y}(x, y) = p_X(x)\, p_Y(y)$

5.3 Independence of Random Variables 2

or $f_{X,Y}(x, y) = f_X(x)\, f_Y(y)$. Lecture 3 uses this factorization in the i.i.d. sampling setup.)

5.3 Independence of Random Variables 3

Useful consequence
If $X \perp\!\!\!\perp Y$, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

This fails without independence. In general $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$, and the difference is exactly the *covariance* — which we introduce on the next slide.

For more than two variables, mutual independence means that their joint distribution factors into the product of their individual distributions.

Where this shows up later. Independence supports the variance rules in §5.5–5.6 and Lecture 3's i.i.d. sampling model. In regression, observations are independent across i; within an observation, $\mathbb{E}[\epsilon_i \mid X_i] = 0$ (Lectures 5–7).

5.4 Covariance

Covariance measures how two random variables *move together*. In §5.3 we saw that $X \perp\!\!\!\perp Y$ implies the product expectation factorizes: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. When that fails, the difference is the covariance.

Definition
The covariance of two random variables X and Y is

$$\mathrm{Cov}(X, Y) \;=\; \mathbb{E}[\,(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\,] \;=\; \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y].$$

The two expressions are equal — expand the product and use linearity, just like the computational variance formula (Lecture 1.7.4).

Notation reminder. $\mathrm{Cov}(\cdot, \cdot)$ takes two random variables as inputs and returns a single number. The number can be positive, negative, or zero.

5.4 Covariance 2

Intuition. Think of $X - \mathbb{E}[X]$ as "how far above (or below) its mean X is" — and similarly for Y. The product is positive when X and Y are both above their means or both below; negative when one is above and the other below. Averaging the product over the distribution tells you which case dominates:

- $\mathrm{Cov}(X, Y) > 0$: above-average X tends to come with above-average Y. They move in the *same* direction.
- $\mathrm{Cov}(X, Y) < 0$: above-average X tends to come with below-average Y. *Opposite* directions.
- $\mathrm{Cov}(X, Y) = 0$: no linear association.

5.4 Covariance 3

Key properties

- Self: $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.
- Symmetric: $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$.
- Independence implies zero covariance: $X \perp\!\!\!\perp Y \;\Rightarrow\; \mathrm{Cov}(X, Y) = 0$.
- Bilinearity: for constants a, b, c, d,

$$\mathrm{Cov}(aX + b,\; cY + d) \;=\; ac \cdot \mathrm{Cov}(X, Y).$$

Adding constants has no effect; scaling multiplies covariance by ac.

Warning. Zero covariance does *not* imply independence. It rules out only *linear* association; for example, $Y = X^2$ can be strongly related to a symmetric X even when $\mathrm{Cov}(X, Y) = 0$.

Later: Lecture 7 uses covariance to interpret the population OLS slope.

5.5 Linear Combinations of Two Random Variables

Let X, Y be random variables and a, b constants. The combination $W = aX + bY$ is itself a random variable. We need its mean and variance.

Key Property — Expectation of $aX + bY$

$$\mathbb{E}[aX + bY] \;=\; a\,\mathbb{E}[X] \;+\; b\,\mathbb{E}[Y].$$

True always, with or without independence. This is just linearity (Lecture 1.7.2) applied twice.

Key Property — Variance of $aX + bY$ (general)
For any X, Y and constants a, b,

$$\mathrm{Var}(aX + bY) \;=\; a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

5.5 Linear Combinations of Two Random Variables 2

Special case — Independent X, Y
If $X \perp\!\!\!\perp Y$, then $\mathrm{Cov}(X, Y) = 0$ and the formula collapses:

$$\mathrm{Var}(aX + bY) \;=\; a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y).$$

The general formula says: variance of a sum has two pieces — each variable's own contribution ($a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)$), plus the interaction ($2ab\;\mathrm{Cov}(X, Y)$). The cross term vanishes exactly when the variables don't co-move.

5.6 General Linear Combination: $\sum a_i X_i$

Now scale up to n random variables. Let $X_1, \ldots, X_n$ be random variables and $a_1, \ldots, a_n$ be constants. Consider

$$W = a_1 X_1 + a_2 X_2 + \cdots + a_n X_n = \sum_{i=1}^{n} a_i X_i.$$

Key Property — Expectation (always)

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i\right] = \sum_{i=1}^{n} a_i \, \mathbb{E}[X_i].$$

No independence assumption needed. Pure linearity.

5.6 General Linear Combination: $\sum a_i X_i$ 2

Key Property — Variance (independent X_i)
If $X_1, \ldots, X_n$ are mutually independent,

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \, \mathrm{Var}(X_i).$$

5.6 General Linear Combination: $\sum a_i X_i$ 3

Special case (constants $a_i = 1$). If $X_1, \ldots, X_n$ are independent with common mean μ and common variance σ^2, then for the simple sum $S_n = X_1 + \cdots + X_n$:

$$\mathbb{E}[S_n] = n\mu, \qquad \mathrm{Var}(S_n) = n\sigma^2.$$

Why these formulas matter. Almost every estimator we build in this course is a linear combination of random variables — the OLS estimator $\hat{\beta}$ (Lecture 6), sample averages and estimators introduced in Lecture 3, any forecast that averages predictions. These two key properties are the toolkit we use to find their mean and variance.

Where this shows up later. Sampling distributions (Lecture 3); standard errors of OLS estimators (Lecture 7); the bias–variance decomposition (Lecture 10).

5.7 Why We Won't Compute Integrals

An honest aside.

Continuous-RV formulas are written with integrals — that's the language of the field, and you'll see those formulas all over textbooks and papers. But in this course:

- We use integration conceptually, as "area under the curve."
- Every numerical answer we need will come from a table or from software.
- You will never be asked to compute an integral on an exam.

Why bother showing the integrals at all? Because we want you to be able to *read* the formulas, recognize what they say geometrically, and trust where the numerical answers come from. The picture is the point; the integral notation is the bookkeeping.

6. The Uniform Distribution

6.1 Motivating Example: Refrigerator Delivery

Setup. The delivery company tells you the refrigerator will arrive sometime between 2:00 pm and 3:00 pm, with no time more likely than any other. Let X be the delivery time in minutes after 2:00. So $X \in [0, 60]$, and "uniformly random" means the density is *flat*.

Question: what is the probability of delivery before 2:20?

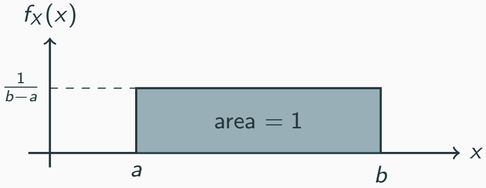
Quick answer (geometric). The total density spans 60 minutes; we want the first 20 of them; so $\mathbb{P} = 20/60 = 1/3$. Read straight off a rectangle. We now formalize.

Definition
A random variable X has the uniform distribution on $[a, b]$, written

$$X \sim U[a, b],$$

if its density is

$$f_X(x) = \begin{cases} \dfrac{1}{b-a} & a \leq x \leq b, \\ 0 & \text{otherwise.} \end{cases}$$



6.3 Mean of the Uniform (Geometric Derivation)

Result

If $X \sim U[a, b]$, then $\mathbb{E}[X] = \dfrac{a+b}{2}$.

The picture is the proof. The density is a symmetric rectangle on $[a, b]$. Its balance point sits at the geometric center of the interval:

$$\mathbb{E}[X] \;=\; \text{center of } [a, b] \;=\; \frac{a+b}{2}.$$

For the algebra-curious (not required): plug $f_X(x) = 1/(b-a)$ into the definition and integrate:

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a} \cdot \left[\frac{x^2}{2}\right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

Same answer. The picture is the real argument.

6.4 Variance of the Uniform

Result

If $X \sim U[a, b]$, then $\mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$.

We state this without deriving it (the derivation is in the appendix slide for the curious).

Intuition. Variance scales as the *square* of the interval length:

- Double the interval (from $[0, 1]$ to $[0, 2]$) and the variance quadruples.
- This is consistent with $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$ from Lecture 1.7.5: stretching the interval by a factor of a multiplies the variance by a^2.

6.5 Worked Example: $X \sim U[2, 12]$

Setup. $X \sim U[2, 12]$. The PDF is a rectangle on $[2, 12]$ of height $1/10$.

Step 1: probabilities by area.

$$\begin{aligned} \mathbb{P}(X \leq 2) &= 0 \quad \text{(no area to the left of 2)}, \\ \mathbb{P}(2 \leq X \leq 10) &= 8 \times \tfrac{1}{10} = 0.8 \quad \text{(rectangle base} \times \text{height)}, \\ \mathbb{P}(X = 3) &= 0 \quad \text{(single point)}. \end{aligned}$$

Step 2: mean and variance.

$$\mathbb{E}[X] = \frac{2 + 12}{2} = 7, \qquad \mathrm{Var}(X) = \frac{(12-2)^2}{12} = \frac{100}{12} \approx 8.33.$$

Interpretation. The distribution is centered at 7 with "typical deviation" $\mathrm{SD}(X) = \sqrt{8.33} \approx 2.89$. Every probability of interest is just an area of a rectangle.

6.A (Appendix) Derivation of Var(Uniform)

For the curious — not required.

Let $X \sim U[a, b]$, $\mu = (a + b)/2$. Use the computational formula:

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2.$$

Compute $\mathbb{E}[X^2]$ by integrating:

$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a}\, dx = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3}.$$

Subtract $\mu^2 = (a + b)^2/4$:

$$\begin{aligned}\mathrm{Var}(X) &= \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} \\ &= \frac{4(a^2 + ab + b^2) - 3(a+b)^2}{12} = \frac{(b-a)^2}{12}.\end{aligned}$$

Done.

7. The Normal Distribution

7.1 Motivating Example: Bottle Filling

Setup. A machine fills soda bottles with some random variation. Suppose the amount X (in ml) is approximately normally distributed with mean $\mu = 590$ and standard deviation $\sigma = 5$.

Question. What is the probability that a random bottle contains more than 600 ml?

- We can't read this off a rectangle (the density isn't flat).
- We will resolve it in §8.5, after we set up the Normal distribution and the standardization trick.

This is the most important distribution in statistics — so we take our time.

7.2 The Normal PDF

Definition
A random variable X has the Normal distribution (also called Gaussian) with parameters μ and σ^2, written $X \sim N(\mu, \sigma^2)$, if its density is

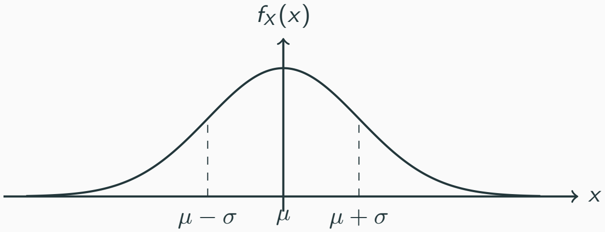
$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right).$$

Notation reminder. μ is the Greek letter "mu"; σ is "sigma." μ is the *mean* and σ^2 is the *variance*; σ alone is the *standard deviation*.

Common pitfall. In the notation $N(\mu, \sigma^2)$, the second argument is variance, not standard deviation. So $N(0, 4)$ means mean 0 and variance 4 (so $\sigma = 2$), not $\sigma = 4$. Software and textbooks vary — always check.

Don't memorize the formula. The constant $1/(\sigma\sqrt{2\pi})$ exists only to make the total area equal to 1. You will never use it numerically.

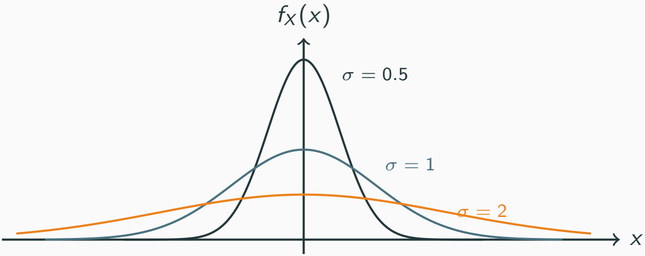
7.3 Shape of the Normal



Four facts to know:

- Symmetric about μ: $f_X(\mu - c) = f_X(\mu + c)$.
- Bell-shaped: peak at μ, drops off in both directions.
- Mean $=$ median $=$ mode all equal μ.
- Defined on *all* of $\mathbb{R}$: the tails approach zero but never reach it.

7.4 μ Shifts, σ Stretches



All three curves have the *same* $\mu = 0$ but different σ. Each one has total area 1.

- Small $\sigma \Rightarrow$ tall and narrow.
- Large $\sigma \Rightarrow$ short and wide.
- μ shifts the curve left/right without changing shape; σ stretches/compresses it without moving the center.

7.5 Mean and Variance of the Normal

Result
If $X \sim N(\mu, \sigma^2)$, then

$$\mathbb{E}[X] = \mu, \qquad \mathrm{Var}(X) = \sigma^2.$$

Why? Because we *defined* the parameters μ and σ^2 that way. They are not derived from a calculation; they are the parameters of the distribution by construction. The symmetric bell shape balances at μ, and the spread parameter σ is exactly the standard deviation.

No derivation to spell out — just know the names match the things they're named after.

7.6 Linear Combinations of Independent Normals

We've already seen (Lecture 1.7.2, §5.5–5.6) that linearity gives us the *mean and variance* of any linear combination of random variables. For Normals, something stronger holds: the shape is preserved too.

Key Property — Normal-specific!
If $X_1, \ldots, X_n$ are independent with $X_i \sim N(\mu_i, \sigma_i^2)$, then any linear combination is itself Normal:

$$\sum_{i=1}^{n} a_i X_i \;\sim\; N\left(\sum_{i=1}^{n} a_i \mu_i,\; \sum_{i=1}^{n} a_i^2 \sigma_i^2\right).$$

This does NOT hold for general distributions. For most distributions, a sum of independent copies has a *different* shape than the originals — e.g., the sum of two independent Uniforms is a *triangular* distribution,

7.6 Linear Combinations of Independent Normals 2

not a Uniform. The Normal is one of the few families closed under linear combinations.

Where this shows up later. OLS coefficients are linear combinations of the response values y_i; if the regression errors are Normal, the coefficient estimates are Normal too — which is exactly why we can build the inference machinery (Z-tests and confidence intervals) of Lecture 7.

7.7 What Is and Isn't "Normal-ish"

Things that are approximately Normal:

- Heights and shoe sizes (in a single sex, single age group).
- SAT and IQ scores (by design).
- Measurement errors and lab noise.
- Sample means of many independent observations (the CLT — Lecture 3).

Things that are *not* Normal:

- Incomes and wealth (right-skewed, heavy upper tail).
- Stock returns (heavy tails, occasional big jumps).
- Reaction times (right-skewed).
- Counts of rare events (Poisson-like, often skewed).

Where this shows up later. The Central Limit Theorem (Lecture 3) explains *why* so many sampling distributions look Normal. Large-sample regression inference uses this idea in Lecture 7.

8. Standardization and the Z-Score

8.1 The Z-Score

Definition
For $X \sim N(\mu, \sigma^2)$, the standardized version of X is

$$Z = \frac{X - \mu}{\sigma}, \qquad Z \sim N(0, 1).$$

Z is called the standard normal and follows $N(0, 1)$ — mean 0, variance 1.

Notation reminder. The letter Z is conventional for a standard Normal random variable.

Intuition.

- Subtracting μ centers the distribution at 0.
- Dividing by σ rescales it to unit spread.
- Same bell shape — just shifted and stretched.

8.2 Why Standardization Works (Derived)

Claim: any linear function of a Normal is again Normal. So $Z = (X - \mu)/\sigma$ is Normal; we just need to find its mean and variance.

Mean. Apply linearity of expectation (Lecture 1.7.2):

$$\mathbb{E}[Z] = \mathbb{E}\left[\frac{X - \mu}{\sigma}\right] = \frac{1}{\sigma}\mathbb{E}[X] - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0.$$

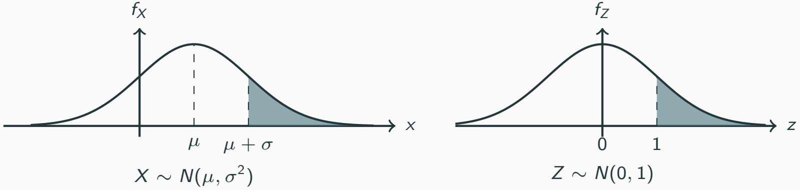
Variance. Apply $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$ (Lecture 1.7.5). Here $a = 1/\sigma$, $b = -\mu/\sigma$:

$$\mathrm{Var}(Z) = \left(\frac{1}{\sigma}\right)^2 \mathrm{Var}(X) = \frac{1}{\sigma^2} \cdot \sigma^2 = 1.$$

So $Z \sim N(0, 1)$. The standardization recipe is just linearity + affine variance rule applied to a Normal.

8.3 Geometric View: Standardization Preserves Area

Shifting and rescaling *doesn't change probabilities* — it just relabels the axis.



Both shaded areas are equal: $\mathbb{P}(X > \mu + \sigma) = \mathbb{P}(Z > 1)$. That's why every Normal computation reduces to a question about Z.

8.4 The Z-Table: Idea and Modern Replacement

The CDF of the standard Normal
The standard-Normal CDF is denoted

$$\Phi(z) \ = \ \mathbb{P}(Z \leq z), \qquad Z \sim N(0,1).$$

Notation reminder. Φ is the Greek capital letter "phi" — the standard symbol for this CDF.

Mini lookup example. A Z-table tells you

$$\Phi(2) \approx 0.977.$$

In words: 97.7% of the standard-Normal density lies to the left of $z = 2$. Equivalently, $\mathbb{P}(Z > 2) \approx 1 - 0.977 = 0.023$.

8.4 The Z-Table: Idea and Modern Replacement 2

A note on Z-tables. In the pre-computer era, statisticians carried printed Z-tables in the back of every textbook — you found $\Phi(2)$ by reading row 2.0, column .00. Older textbooks (and the first probability course many of you took) lean on these tables heavily.

Today we just ask software to compute $\Phi(z)$ to as many digits as we want. The table is a relic. What you need to keep is the *idea*: $\Phi(z)$ is an area you can look up.

8.5 Worked Example: Bottle Filling, Resolved

Setup. $X \sim N(\mu = 590, \sigma^2 = 25)$, so $\sigma = 5$. Find $\mathbb{P}(X > 600)$.

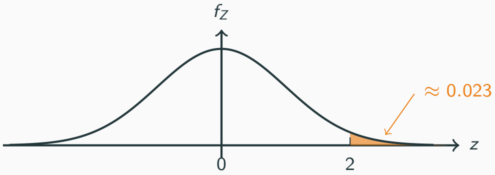
Step 1: standardize.

$$\mathbb{P}(X > 600) \;=\; \mathbb{P}\left(\frac{X - 590}{5} > \frac{600 - 590}{5}\right) \;=\; \mathbb{P}(Z > 2).$$

Step 2: use Φ. The right tail beyond $z = 2$ is the complement of $\Phi(2)$:

$$\mathbb{P}(Z > 2) = 1 - \Phi(2) \approx 1 - 0.977 = 0.023.$$

8.5 Worked Example: Bottle Filling, Resolved 2



Interpretation. About 2.3% of bottles overfill past 600 ml.

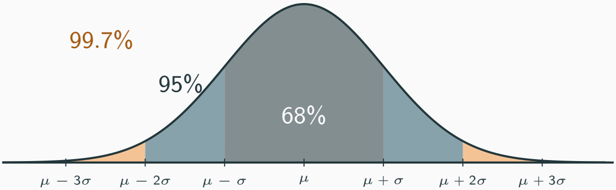
Where this shows up later. Standardization is the engine behind every test statistic and every confidence interval in Lectures 4 and 7. (We use Z throughout — see Lecture 4 §3 for why we skip the t-distribution.)

9. The Empirical Rule (68–95–99.7)

9.1 The Empirical Rule

Key result
If $X \sim N(\mu, \sigma^2)$, then

$$\mathbb{P}(\mu - \sigma < X < \mu + \sigma) \approx 0.68,$$
$$\mathbb{P}(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95,$$
$$\mathbb{P}(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997.$$



9.2 Where the Numbers Come From

Not magic — just Z-table arithmetic.

By symmetry of the standard Normal,

$$\mathbb{P}(-k < Z < k) = 2\,\Phi(k) - 1.$$

Plug in $k = 1, 2, 3$ with $\Phi(1) \approx 0.841$, $\Phi(2) \approx 0.977$, $\Phi(3) \approx 0.9987$:

$$\begin{aligned}
\mathbb{P}(-1 < Z < 1) &= 2(0.841) - 1 \approx 0.683, \\
\mathbb{P}(-2 < Z < 2) &= 2(0.977) - 1 \approx 0.954, \\
\mathbb{P}(-3 < Z < 3) &= 2(0.9987) - 1 \approx 0.997.
\end{aligned}$$

And the empirical rule for a general $N(\mu, \sigma^2)$ follows because $\mu - k\sigma < X < \mu + k\sigma$ is the same event as $-k < Z < k$ after standardizing.

9.3 Worked Example: SAT Scores

Setup. SAT scores are approximately Normal with mean $\mu = 1000$ and standard deviation $\sigma = 200$. So $X \sim N(1000, 200^2)$.

Step 1: apply the empirical rule.

$$\mathbb{P}(800 < X < 1200) \approx 0.68 \quad (\mu \pm \sigma),$$
$$\mathbb{P}(600 < X < 1400) \approx 0.95 \quad (\mu \pm 2\sigma),$$
$$\mathbb{P}(400 < X < 1600) \approx 0.997 \quad (\mu \pm 3\sigma).$$

Step 2: what fraction scores above 1400? Score 1400 is $\mu + 2\sigma$. By the empirical rule, 5% of scores fall outside $[\mu - 2\sigma, \mu + 2\sigma]$; by symmetry, 2.5% are above and 2.5% are below. So $\mathbb{P}(X > 1400) \approx 0.025$.

Interpretation. A mental estimate, no calculator. About 1 student in 40 scores above 1400.

Where this shows up later. The 95% band ($\mu \pm 2\sigma$) is the prototype for the 95% confidence intervals you will build in Lecture 4.

10. The Regression Function: Conditional Expectation

10.1 From Events to Random Variables

In Lecture 1.4 we learned how to condition one *event* on another: $\mathbb{P}(A \mid B)$, the probability of A given that B happened.

We now extend the same idea to two *random variables* (X, Y). Suppose we observe many pairs of people's (X, Y) — for example, X = years of education and Y = annual income. A natural question:

What is the *average* value of Y, among observations where X took a specific value x?

That average is the *conditional expectation* of Y given X, written $\mathbb{E}[Y \mid X = x]$. It generalizes "conditional probability" from events to random variables.

10.2 Conditional Expectation

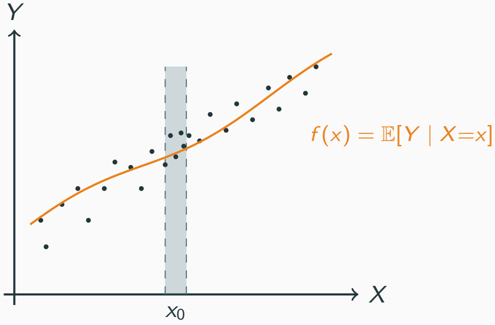
Definition
The conditional expectation of Y given $X = x$, written $\mathbb{E}[Y \mid X = x]$, is the mean of Y *among observations where* X *equals* x.

Varying x traces out a function of x. Call it f:

$$f(x) \equiv \mathbb{E}[Y \mid X = x].$$

Notation reminder. The bar $\mid$ is read "given," the same as in Lecture 1.4. The new ingredient is that what's on either side of the bar is now a random variable, not a single event.

10.2 Conditional Expectation 2



10.3 Worked Example: Income Given Education

Setup. Let Y = annual income and X = years of education in a large population of workers. Suppose:

- Among workers with $X = 12$ (high school graduates), mean income is \$45,000.
- Among workers with $X = 16$ (college graduates), mean income is \$70,000.

Step 1: read off conditional expectations.

$$\mathbb{E}[Y \mid X = 12] = 45{,}000, \qquad \mathbb{E}[Y \mid X = 16] = 70{,}000.$$

Step 2: as x varies, you trace a curve. The mapping

$$x \longmapsto \mathbb{E}[Y \mid X = x]$$

10.3 Worked Example: Income Given Education 2

is itself a function. We call it $f(x)$.

Interpretation. $f(x) = \mathbb{E}[Y \mid X = x]$ is the ideal predictor of Y given X. Among all possible ways to predict Y from X, this curve has the smallest expected squared error. It is what regression tries to learn.

Where this shows up later. Lecture 5 introduces this $f(x)$ as "the relationship between X and Y." OLS (Lecture 6) approximates it by a straight line; later lectures (Lectures 10–14) try to approximate it with richer models.

11. Summary

11.1 Take Away — Foundations

Continuous RVs in one slide

- For continuous X: probabilities are areas under the density f_X; the height $f_X(x)$ is not a probability.
- $\mathbb{P}(X = x) = 0$ for every fixed x; only ranges have positive probability.
- PDF f_X satisfies $f_X \geq 0$ and total area $= 1$.
- CDF $F_X(x) = \mathbb{P}(X \leq x)$, and $\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a)$.
- Mean is the balance point of the density; $\mathbb{E}[X] = \int x f_X(x)\, dx$.

11.2 Take Away — Algebraic Key Properties

Exam targets

- Linearity (always): $\mathbb{E}[\sum_i a_i X_i] = \sum_i a_i \mathbb{E}[X_i]$.
- Covariance: $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$; symmetric; $\mathrm{Cov}(aX + b, cY + d) = ac\, \mathrm{Cov}(X, Y)$; $X \perp\!\!\!\perp Y \Rightarrow \mathrm{Cov}(X, Y) = 0$ (converse fails — zero covariance does not mean independent).
- Variance of $aX + bY$ (general): $\mathrm{Var}(aX+bY) = a^2\, \mathrm{Var}(X) + b^2\, \mathrm{Var}(Y) + 2ab\, \mathrm{Cov}(X, Y)$.
- Variance of $\sum_i a_i X_i$ (mutually independent): $\mathrm{Var}(\sum_i a_i X_i) = \sum_i a_i^2\, \mathrm{Var}(X_i)$.

Uniform, Normal, and friends

- $X \sim U[a, b]$: $\mathbb{E}[X] = (a + b)/2$, $\text{Var}(X) = (b - a)^2/12$.
- $X \sim N(\mu, \sigma^2)$: bell-shaped, symmetric, mean μ, variance σ^2 (second argument is variance, not SD).
- Standardize: $Z = (X - \mu)/\sigma \sim N(0, 1)$. Any linear function of a Normal is Normal.
- Linear combinations of independent Normals are again Normal (Normal-specific — does NOT hold for general distributions).
- Empirical rule: $\approx 68/95/99.7\%$ of mass within $\pm 1, 2, 3$ standard deviations.
- $f(x) = \mathbb{E}[Y \mid X = x]$ is the ideal predictor of Y from X — the regression target.

11.4 Where Each Tool Reappears in This Course

Forward pointers

- Normal & Z-scores — sampling distributions and the formal CLT (Lecture 3); Z-statistics for inference (Lectures 4 and 7).
- Standardization — confidence intervals (Lecture 4); hypothesis tests on regression coefficients (Lecture 7).
- Empirical rule — the 95% prototype for confidence intervals (Lecture 4).
- Variance rules + covariance — the population OLS slope and coefficient standard errors (Lecture 7).
- $\mathbb{E}[Y \mid X]$ — regression function $f(x)$ in Lecture 5; OLS line in Lecture 6; richer approximations in Lectures 10–14.

Next time: estimators, sampling distributions, the CLT.